A SHORT COURSE. A USEFUL REFERENCE.

Peptides, made simple.

Understand the science.
Learn how to compare.

A guided introduction to the materials in the NXT LVL peptide catalog, the evidence behind them and the questions to ask before planning research.



YOUR LEARNING PATH

01 Start with the basics 02-03
What peptides are, how evidence works and what labels mean.

02 Meet the materials 04-10
28 profiles covering the 31 listings in this catalog snapshot.

03 Understand combinations 11
Why products are paired and what a bundle does not prove.

04 Work out quantities 12
Vial math, a worked example and the right units.

05 Learn preparation principles 13-14
Concentration, handling, storage and a final review.

Tap a lesson or product name to jump.

For laboratory research education

This course explains research materials. It does not provide personal treatment, injection or cycle instructions. For health decisions, consult a licensed clinician.

LESSON 01 / FOUNDATIONS

Start with the big picture

Learning goal: understand what a peptide can signal, and what a vial label tells you.

Peptides are short chains of amino acids.

Amino acids are building blocks also found in proteins. Some peptides act as signals: they interact with a target in or on a cell and can change what that cell does. Different sequences can have very different effects.

MOLECULE	TARGET	RESPONSE
The material being studied	A receptor or cell process	A measurable change

A response is not automatically a benefit. A change in a hormone level, for example, does not by itself prove better recovery, longer life or improved health.

Read the amount before the name.

On a label	What it tells you
10 mg	Mass of material listed in the vial. It does not mean 10 mL or one treatment.
10 mL	Liquid volume or container capacity. It does not reveal the amount in mg.
5,000 IU	Biological activity for that particular material. IU and mg are not interchangeable.

Remember this

The molecule, amount, concentration and container size describe different things. Learn them separately before doing any quantity calculation.

LESSON 01 / EVIDENCE

Know what the evidence says

Learning goal: separate a research idea, a human finding and an approved treatment.

Evidence label	How to read it
Early research	Cells or animals were studied. Human benefit and safety are not established.
Limited human data	People have been studied, but the research does not establish routine use.
Approved medicine exists	A regulated formulation has specific uses. The catalog research vial is not that medicine.
Identity needs checking	The exact molecule or each blend ingredient must be confirmed before evaluating it.

Use four checks when comparing a study.

Same material? Check the sequence, form and ingredient amounts.

Same route? A skin or nasal study does not establish injection safety.

Same setting? Cells, mice and patients answer different questions.

Same outcome? A lab marker is not the same as a clinical benefit.

What a lot report can tell you

A certificate of analysis (COA) describes tests on a sample. Match the product and lot, then check which tests were actually performed. Identity, amount, purity, sterility and bacterial toxins are separate questions. A high purity number does not answer all of them.

Before ordering: confirm the exact identity, amount per vial, matching lot documentation, storage instructions and solvent required by your method. Ask for missing information.

LESSON 02 / MATERIALS 1 OF 7

Tissue repair and blends

Amounts are listed per vial, not doses. Tap a product name to see its current listing.

BPC-157

10 mg | Mostly early research

What it is: A small experimental peptide studied in tissue and digestive-system models.
Research focus: Researchers investigate repair responses in tendon, muscle and gut tissue.
Know this: Most evidence is from cells or animals. Reliable human healing benefits and safety are not established.

TB-500

10 mg | Mostly early research

What it is: A name used for material related to thymosin beta-4, a protein involved in cell movement.
Research focus: Studied in tissue-repair and wound-healing research.
Know this: Confirm the exact molecule: the TB-500 fragment and full-length thymosin beta-4 are different. Human wound-healing benefit is unestablished.

TB-500 + BPC-157

10 mg TOTAL blend | Formula details missing

What it is: A single vial containing two named materials.
Research focus: The combination brings together two tissue-research interests.
Know this: The listing does not give each ingredient's amount. Do not assume 10 mg of each, a 50/50 split, or better results together.

GLOW

70 mg TOTAL blend | Formula details missing

What it is: A blend name, rather than one identifiable peptide.
Research focus: Its research purpose depends on the actual ingredients.
Know this: The full formula is not published in the reviewed listing. Get ingredient names and individual amounts before evaluating or ordering it.

TAKEAWAY A blend name or two ingredients do not establish better results together. Confirm the complete formula.

LESSON 02 / MATERIALS 2 OF 7

Skin, inflammation and fuel

Amounts are listed per vial, not doses. Tap a product name to see its current listing.

GHK-Cu

Other size listing >

50 mg or 100 mg | Mostly early research

What it is: A short peptide attached to copper.
Research focus: Researchers study collagen production and how tissue rebuilds.
Know this: Cell, animal or skin-applied findings do not prove benefits from injecting it. The 100 mg vial contains twice the listed material of the 50 mg vial.

KPV

10 mg | Mostly early research

What it is: A very short peptide studied for inflammation-related signals.
Research focus: Relevant to laboratory questions about inflammation in tissue and gut models.
Know this: Human benefit and safety are not established. It is not a proven treatment for gut or skin problems.

MOTS-c

10 mg | Mostly early research

What it is: A peptide linked to mitochondria, the energy-processing parts of cells.
Research focus: Studied for how cells handle fuel and respond to insulin.
Know this: Promising animal findings do not establish human energy, weight-loss or longevity benefits.

AOD-9604

10 mg | Human benefit unproven

What it is: A modified fragment of growth hormone, studied for fat metabolism.
Research focus: Investigated as a possible way to affect fat breakdown.
Know this: Human studies did not establish the intended weight-loss benefit. It is not a proven fat-loss or spot-reduction product.

TAKEAWAY A promising cell or animal result is a starting point for research, not proof of a human benefit.

LESSON 02 / MATERIALS 3 OF 7

Appetite, weight and energy

Amounts are listed per vial, not doses. Tap a product name to see its current listing.

Semaglutide

10 mg | Approved medicine exists

What it is: Activates GLP-1, a signal involved in appetite and blood-sugar control.
Research focus: Approved prescription formulations have demonstrated weight-management benefits in eligible patients.
Know this: The research vial is not the approved medicine. Risks include severe digestive effects, pancreas and gallbladder problems, and thyroid-related restrictions.

Tirzepatide

10 mg | Approved medicine exists

What it is: Activates two signals: GIP and GLP-1.
Research focus: Approved prescription formulations have demonstrated weight-management benefits in eligible patients.
Know this: A research vial is not the prescription product. Serious risks and thyroid-related restrictions apply; Zepbound labeling advises against adding another GLP-1 medicine.

Retatrutide

Other size listing >

10 mg or 20 mg | Human trials; investigational

What it is: Activates three signals: GLP-1, GIP and glucagon.
Research focus: Clinical trials study effects on body weight and metabolism.
Know this: It remains investigational. Trial success does not establish the quality or safety of research vials. Digestive effects and increased heart rate are concerns.

SLU-PP-332

10 mg | Mostly early research

What it is: An experimental small molecule. It is not a peptide.
Research focus: Researchers study energy use and endurance-related changes in mice.
Know this: Human benefit and safety are unestablished. It is not a proven exercise replacement, and a peptide-water mixing recipe cannot be assumed suitable.

TAKEAWAY Different product names can act on overlapping signals. More products do not automatically mean more benefit.

LESSON 02 / MATERIALS 4 OF 7

Growth-hormone signals

Amounts are listed per vial, not doses. Tap a product name to see its current listing.

Tesamorelin

10 mg | Approved medicine exists

What it is: Prompts the pituitary gland to release growth hormone.
Research focus: Approved Egrifta treats excess abdominal fat in adults with HIV-associated lipodystrophy, a specific fat-distribution condition.
Know this: That evidence does not make the research vial a general fat-loss product. Blood sugar, IGF-1 and fluid retention are concerns.

Sermorelin

10 mg | Historical medical use

What it is: A short signal that prompts growth-hormone release.
Research focus: Useful for understanding the body's growth-hormone signaling system; prescription products had historical growth-related uses.
Know this: That history does not prove adult anti-aging or recovery benefits. A research vial needs its own identity and quality documentation.

CJC-1295 with DAC

10 mg | Limited human evidence

What it is: A growth-hormone-releasing signal with a modification that prolongs its action.
Research focus: Small human studies found longer-lasting increases in growth hormone and IGF-1.
Know this: Higher hormone measurements are not proof of more muscle, fat loss or longevity. Clinical safety data are limited.

CJC-1295 without DAC

10 mg | Identity needs confirmation

What it is: A non-DAC product often marketed as modified GRF 1-29.
Research focus: Discussed in research on signals that release growth hormone.
Know this: Check the exact identity. It is different from the longer-acting DAC form, so the DAC study results cannot simply be applied to it.

TAKEAWAY GH means growth hormone. IGF-1 is a growth-related signal influenced by GH. Several products affect this same system.

LESSON 02 / MATERIALS 5 OF 7

Growth, sleep and anxiety

Amounts are listed per vial, not doses. Tap a product name to see its current listing.

Ipamorelin

10 mg | Limited human evidence

What it is: Stimulates growth-hormone release through a different receptor from the CJC/sermorelin pathway.
Research focus: Researchers examine how that separate signaling route affects hormone release.
Know this: A plausible mechanism does not prove better sleep, muscle gain or recovery. Human safety information is limited and depends on the route studied.

IGF-1 LR3

1 mg | Mostly early research

What it is: A modified version of IGF-1, a growth-related signal.
Research focus: Used in experimental work on cell and tissue growth.
Know this: It is different from prescription mecasermin. Human muscle-building benefit is unproven; growth-related and blood-sugar effects require careful evaluation.

DSIP

10 mg | Limited human evidence

What it is: Short for delta sleep-inducing peptide; also called emideltide.
Research focus: Older studies investigated sleep and withdrawal-related outcomes.
Know this: The name sounds like a promise, but effectiveness for insomnia is not established. It is not a validated sleep stack ingredient.

Selank

11 mg | Limited human evidence

What it is: An experimental peptide investigated for anxiety-related effects.
Research focus: Small human research has reported changes in anxiety symptoms.
Know this: Evidence is limited. Those findings do not establish a general anxiety treatment or the safety of injecting a research vial.

TAKEAWAY An effect on a pathway or a promising name does not establish a useful treatment.

LESSON 02 / MATERIALS 6 OF 7

Brain, pigment and desire

Amounts are listed per vial, not doses. Tap a product name to see its current listing.

Semax

11 mg | Limited human evidence

What it is: An experimental peptide investigated for effects in the nervous system.
Research focus: Research has explored reduced blood flow to the brain, migraine and nerve-pain conditions.
Know this: Effectiveness was not established in the cited review. Most human reports involved nasal use, which does not validate injections or a focus-enhancement claim.

Melanotan-1

10 mg | Approved medicine exists

What it is: Usually refers to afamelanotide, which affects skin pigment.
Research focus: An approved implant improves light tolerance in adults with a rare light-sensitivity disorder called erythropoietic protoporphyria.
Know this: The research vial is not that implant or an established cosmetic-tanning treatment. Pigment changes need attention; sun protection remains necessary.

Melanotan-2

10 mg | Limited human evidence

What it is: An experimental material affecting pigment and other melanocortin signals.
Research focus: A small early human study observed pigmentation and sexual effects.
Know this: Long-term safety is uncertain. Nausea and prolonged erections are concerns; adding PT-141 is not a validated combination.

PT-141

10 mg | Approved medicine exists

What it is: Also called bremelanotide; acts on melanocortin signals involved in sexual desire.
Research focus: Approved Vyleesi treats a specific low-desire disorder in certain premenopausal women.
Know this: This is not proof of general libido or male-performance benefits. Nausea, blood-pressure changes and heart-related restrictions matter.

TAKEAWAY Evidence depends on the exact route, formulation and people studied. Those details matter.

LESSON 02 / MATERIALS 7 OF 7

Hormones and cell chemistry

Amounts are listed per vial, not doses. Tap a product name to see its current listing.

HCG

Other size listing >

5,000 IU or 10,000 IU | Approved medicine exists

What it is: A reproductive hormone that signals the ovaries or testes.
Research focus: Prescription products have specific fertility and hormone-deficiency uses.
Know this: The research listing is not a treatment plan. Excess hormonal stimulation can be harmful; HCG is not a weight-loss aid. IU measure biological activity.

Glutathione

1.5 g = 1,500 mg; listed 10 mL vial | Route matters

What it is: A three-amino-acid compound involved in the cell's antioxidant defenses.
Research focus: Studied for oxidative stress; some oral studies measured changes in skin properties.
Know this: Oral findings do not prove injectable detox or recovery benefits. Contamination has caused serious reactions; vial capacity does not tell you concentration.

NAD+

500 mg | Limited human evidence

What it is: A helper molecule involved in cellular energy reactions. It is not a peptide.
Research focus: Small infusion research measured how the body processes NAD-related compounds.
Know this: A normal role in cells does not prove that supplementation improves energy or longevity. Injectable quality and contamination are important concerns.

Bacteriostatic Water

10 mL | Laboratory supply

What it is: Water with a preservative, used as a mixing liquid when the exact product or method specifies it.
Research focus: Its role is preparation, rather than a biological benefit.
Know this: It is not a universal solvent or a sterilizer. Do not automatically add it to every order; confirm compatibility and the required solvent first.

TAKEAWAY NAD+ is not a peptide; water is a supply. A shared catalog does not make the materials interchangeable.

LESSON 03 / COMBINATIONS

Understand the pairings

Learning goal: distinguish related research interests from proven combined benefits.

A package is a purchase bundle. A blend puts ingredients in one vial. A study of either ingredient alone does not prove that combining them is effective or safe.

Package / separate vials	Why the pairing is discussed
BPC-157 10 mg + TB-500 10 mg	Both relate to tissue-response research. Faster healing together has not been established.
CJC-1295 without DAC 10 mg + Ipamorelin 10 mg	Different receptor routes can affect GH release. The mechanism explains the idea, not a proven combined benefit.
Tesamorelin 10 mg + Ipamorelin 10 mg	Both affect GH release. Results for approved tesamorelin do not validate this combination.
Sermorelin 10 mg + Ipamorelin 10 mg	Different signaling routes, with overlap in the GH system. A defined experiment must justify the pairing.
BPC-157 10 mg + KPV 10 mg	Tissue-response and inflammation research overlap. Better human outcomes together are unestablished.
GHK-Cu 50 mg + GHK-Cu 100 mg	The same material in two quantities: 150 mg total. This is additional supply, not two complementary mechanisms.

Each listed package contains **one vial of each item**: two separate vials. A package is not a complete cycle or evidence of same-vial mixing compatibility.

Overlap matters

Semaglutide, tirzepatide and retatrutide overlap in GLP-1 activity. Multiple GH products and PT-141/Melanotan-2 also overlap in their effects. These are not established combinations to self-select.

LESSON 04 / QUANTITY

Work out the vial count

Learning goal: calculate research inventory from a documented total requirement.

The calculation

Vials needed = total amount required / amount per vial
Round up to the next whole vial. Use the same unit on both sides.

A worked example

A laboratory plan requires **24 mg** in total. Each vial is labeled **10 mg**.
24 / 10 = 2.4, so the inventory requirement is **3 vials**.
This is an invented purchasing example, not an amount to administer.

Example total	Vial amount	Whole vials
44 mg	11 mg	4
1.6 mg	1 mg	2
120 mg	50 mg	3
120 mg	100 mg	2
1,200 mg	500 mg	3
12,000 IU	5,000 IU	3

What changes the answer?

Include controls, repeats and a justified handling allowance in the study total. Check usable life after preparation: a bigger vial may leave more unused material. Calculate each blend ingredient separately; you need its individual amount first.

Cycle length alone is not enough. Weeks do not specify total material. Most research products here have no established human cycle. A documented research protocol or licensed prescriber must supply the appropriate plan; this table does not.

LESSON 05 / PREPARATION

Understand reconstitution

Learning goal: understand concentration and why preparation is product specific.

Reconstitution means adding the specified liquid to a dry material. The right solvent, volume, handling method and storage conditions depend on the exact material and written procedure.

Dilution changes concentration

Concentration (mg/mL) = mass (mg) / final volume (mL)
If dilution is complete with no losses, adding liquid changes concentration; it does not create more peptide.

Same mass	Final volume	Concentration
10 mg	1 mL	10 mg/mL
10 mg	2 mL	5 mg/mL
10 mg	5 mL	2 mg/mL

Arithmetic examples only. These are not recommended mixing volumes. Final volume can differ from liquid added; solvent compatibility and container capacity still need checking.

Follow the preparation record.

1. Confirm the identity, lot, amount, solvent and written method.
2. Use the specified laboratory setup and measuring procedure.
3. Check expected appearance; investigate unexplained changes.
4. Label concentration, date, storage and validated use-by time.

Keep this distinction: bacteriostatic water does not sterilize contaminated powder or remove bacterial toxins. Clear liquid does not prove suitability. There is no universal storage or 28-day rule for every research preparation.

COURSE REVIEW / PUT IT TOGETHER

Your one-page recap

Use this page when you need a quick reminder or want to check your understanding.

If your question concerns...	Start with
Repair and wound responses	Tissue profiles, page 4
Skin, inflammation and cell fuel	Material profiles, page 5
Appetite, weight and energy	Metabolic profiles, page 6
Growth-hormone signaling	GH and growth profiles, pages 7-8
Sleep, anxiety, brain or pigment	Signal profiles, pages 8-9
HCG, glutathione, NAD+ or water	Related materials, page 10

Check your understanding

A vial says 10 mg. Is that a dose?
No. It is the listed material amount.

Does a promising mouse study prove a human benefit?
No. The population and outcome must match the claim.

Do two products in a package belong in one vial?
No. A bundle does not establish mixing compatibility.

Does a 99% purity result prove sterility?
No. Chemical purity and microbial testing are different.

A study needs 24 mg from 10 mg vials. How many?
Three vials, before any additional justified allowance.

Ready to compare? Review the exact product, its evidence and lot documents, then calculate the quantity your research requires.

Open catalog / Review lot reports / Compare packages